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RESEARCH ARTICLE

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An integrated structural health monitoring system for determining local/global responses of historic masonry buildings

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Summary

In this study, an integrated structural health monitoring system was developed and applied to a masonry heritage building under rehabilitation, where diverse structural changes were made. The late 19th century building had over the years undergone significant deterioration, which brought about the appearance of serious pathological processes affecting the building's stability. In 2012, concerns over the severe cracking phenomena affecting the structure prompted the decision to monitor the structure and to undertake strengthening interventions. A global structural monitoring system was applied to the church to identify the occurrence of possible damage mechanisms and monitoring the intervention processes. The monitoring system was based on the integration of subsystems, which were implemented as and when necessary. Traditional sensors consisting of crackmeters, servo-inclinometers, cable extension sensors, displacement transducers, wind vanes, and temperature sensors (thermo-hygrometer and thermocouples) were installed on the building. Moreover, novel measuring approaches using stress sensors and pressure cells were used. Furthermore, manual monitoring elements were installed, which provided measurements for contrasting with those of the continuous sensors, as well as a greater number of monitoring points. Thus, the structural response was measured using 67 monitoring elements. The long-term (5 years) monitoring results indicated that the interventions were carried out without the occurrence of any irregularity. A good relation was confirmed between the manual and continuous measurements. Moreover, the results have shown the capability of the proposed integrated system to evaluate the structural behaviour during all the phases of the interventions.

KEYWORDS

controlled interventions, heritage buildings, masonry, stress sensors, structural health monitoring, traditional sensors

1 | INTRODUCTION

Architectural heritage is an irreplaceable expression of the intellectual richness and cultural diversity of modern societies, and so structural analysis and safety assessment of historical constructions are issues of great importance.[1]

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Therefore, societies have promoted the drafting of specific legislation[2] aimed at protecting and preserving their historical and cultural heritage.

The study of historic buildings is a topic that has been focused on by several authors involved in the preservation of architectural heritage,[3,4] and it requires a multidisciplinary approach involving a variety of specialist expertise and techniques. As the evaluation of the conservation state of historic buildings using destructive techniques should be avoided, wherever possible, to preserve the integrity of the cultural heritage, the development of non-destructive[5,6] and non-contact[7] techniques becomes of crucial importance.

In this context, Structural Health Monitoring (SHM) systems can improve the safety and reliability of structures, reduce maintenance costs, predict failures, and extend service life.[8–10] For these reasons, SHM,[2,11] which is defined as detecting changes to the material or geometric properties of structures, is crucial for building safety,[12,13] being one of the key components of engineering.

According to previous studies,[12,14,15] SHM makes it possible to identify the presence of damage and measure damage evolution. It has received considerable attention and has successfully replaced traditional visual inspection for damage detection and maintenance of structures subjected to various loadings.[16,17] In fact, a SHM system can assist in evaluating the structural behaviour during all the phases of the interventions, and it allows the design solutions to be modified during the strengthening procedure.[18,19] Consequently, these methodologies are potential tools for the identification, localization, and quantification of damage in constructions, as well as proving useful in the intervention follow-up process.

Many different instruments are used in structural monitoring. The selection of the SHM technique to be adopted for each specific application depends on several variables including costs, aims and scope, possible risks, importance of the building, and the available information.[20] Normally, traditional sensors[19–21] for recording the variations of parameters such as crack openings, deformations, or thermal variations are employed in the field of SHM, but new measuring approaches are also being included (e.g., strain sensors, accelerometers, fibre optic sensors, and terrestrial laser scanning[7,17,22–24]). Nevertheless, these devices must be adaptable to the specific circumstances arising in each study of a historic building, with the aim of exhaustively monitoring the behaviour of its structural elements.

With this background, the aim of this research is to present a methodology for the integrated SHM system applied to a historic masonry building during the diagnostic phase and the posterior intervention process. To do so, specialized hardware and software was developed. The flexibility of the SHM system envisaged enabled the integration of different individual subsystems in order to monitor each of the actuations carried out on the building.

The monitoring system was installed in the Church of the Major Seminary of Comillas (Spain). This article reports the methodology followed and a systematic analysis of the recorded data from October 2012 to September 2017 (5 years) by the sensors installed on the church. The structural responses were measured using 26 conventional continuous monitoring sensors (crackmeters, servo-inclinometers, cable extension sensors, displacement transducers, thermo-hygrometers, thermocouples, anemometer, and wind vane), which were complemented with a pressure cell, a frequently used sensor in the scientific community, but which was applied for the first time in a historic building, and four stress sensors designed by our research group. Additionally, 36 manual control points were installed on the church to enable discrete monitoring of the aperture/closure of cracks and inclinations. On the basis of the measured global and local responses of the structure during the monitored period, the SHM system applied ensured the safety and stability of interventions on the historic building.

Consequently, the paper is organized as follows: Section 2 presents the methodology followed. Section 3 describes its application to a historic masonry building: structural problems and interventions carried out on the building; integration of systems of electronic instrumentation, data acquisition, and software development; and actuation protocols. The evolution of the measurements collected from the integrated SHM is discussed in Section 4. Finally, Section 5 presents the conclusions of the paper.

2 | PROPOSED METHODOLOGY

In the process of preserving ancient buildings, engineers should act according to a step-by-step methodology organized in stages similar to those used in medicine[2,25]: condition survey (anamnesis), identification of the causes triggering damage and decay (diagnosis), choice of the remedial measures (therapy), and monitoring of the efficiency of the interventions (control).

Consequently, the first steps correspond to collection of data about the structure, including historical and geometrical information, damage identification, mechanical characterization of materials, and numerical model analysis.[26] Additionally, a global monitoring system was contemplated with the aim of overseeing the structure before, during, and after the intervention process.

An integrated SHM system was designed with the aim of confirming the starting hypothesis and providing real-time feedback during the rehabilitation work on the church. Thus, the real-time conditions could be known, and warnings reported if anomalous situations arose related to possible occurrence of damage, so enabling decisions to be taken about the repair and/or reinforcement actions.

The option chosen was a global system enabling the integration of new subsystems (set of sensors) depending on the requirements of the conditions of the intervention. The choice and design in each case was tailored according to the necessities, also taking into account the following factors:

- The scale of the monitoring of the structure, at local level or in specific structural elements that enabled the interpretation of the global behaviour of the structure.
- Magnitudes to be measured and number of devices.
- Frequency of the readings, taking into consideration the need for supervision of the structure; as well as the data gathering system chosen (manual or continuous).
- Systems of data acquisition, selection of the ideal sensors, and zones to be monitored in the case-study, among others.

The component elements of the system envisaged for the investigation are shown in Figure 1. It consists of data acquisition, through sensors and acquisition subsystems (1); data management and storage, on-site (2); data transmission (3), permitting the remote transfer of information for its processing (4). The system designed also incorporated actuation protocols that systematically evaluated the sensor records, automatically defining alerts if the monitored magnitudes exceeded prefixed limits (5). The next element, diagnosis (6), can be considered one of the most important components of the system. It involved, as well as the interpretation of the data collected and processed, the analysis of the structure's response and the verification of whether damage or deterioration had occurred. That is, diagnosis converts the data obtained from the monitoring into useful information about the condition and behaviour of the structure and, as a consequence, enables decisions to be taken and inform the interested parties about any event occurring.

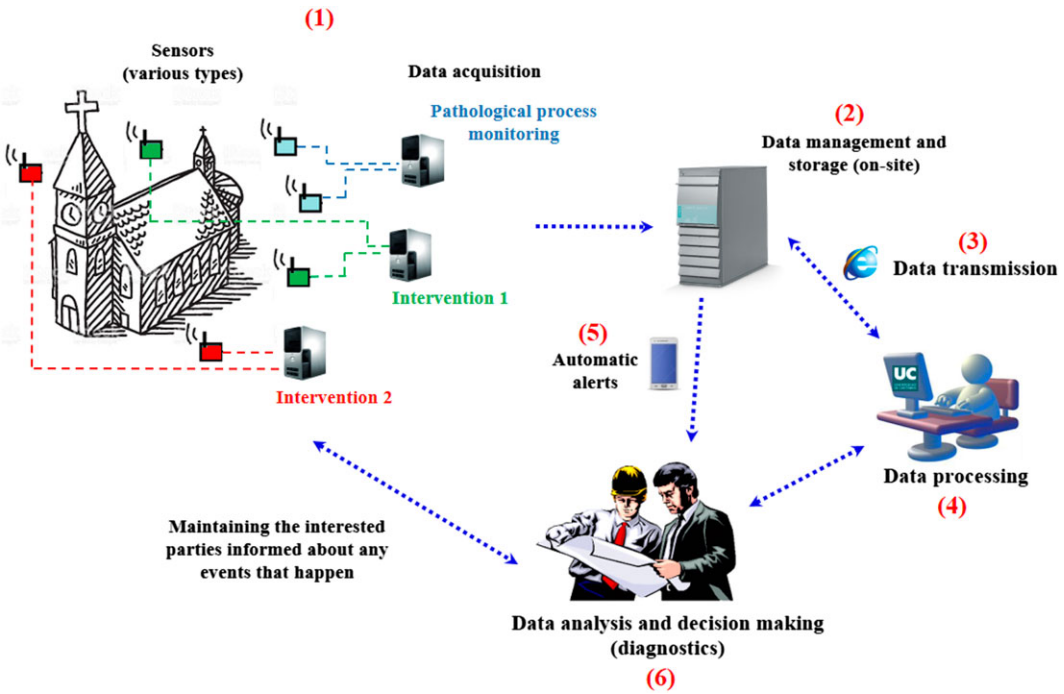


FIGURE 1 Scheme of the component elements of the integrated structural monitoring system envisaged

The integrated SHM system was used in a historic building before, during, and after the intervention process. Five years of monitoring the building enabled the existing pathological structural processes to be monitored, along with the execution process and the efficiency of the interventions. It was a representative case study for the intensive monitoring of a structure; 67 monitoring elements were employed (31 continuous sensors and 36 manual control points), and structural responses (e.g., inclination, displacement, stress, and aperture/closure of cracks) and environmental factors (e.g., environmental temperature and humidity, surface temperature, wind speed, and direction) were monitored during the interventions. Although the whole intervention process was monitored, the system implemented in the building was grouped into different subsystems depending on the actuations developed during the rehabilitation work.

3 | APPLICATION TO A HISTORIC MASONRY BUILDING

3.1 | Description of the building under investigation

3.1.1 | General information about the building

A real-time automatic structural response monitoring system was applied to the Church of the Major Seminary of Comillas, Spain, from the stage of assessment of the pathological structural processes to the stage of intervention. The construction is a late 19th century Modernist building, which is catalogued as a Historic-Artistic Monument,[27] being one of the region's heritage elements of greatest historic-artistic and economic value. The full built complex has a rectangular floor area of 100 × 65 m. It is made up of a church that occupies the centre of the complex and two cloisters, of approximately 35 × 25 m, around it.

The church itself (Figure 2) is made up of a main body, with a central nave, two lateral ones (side chapels), with an apse to the south and the narthex to the north. The principal body is composed of principal transversal arches in the central nave, counteracted by buttresses that separate the lateral chapels. Parallel to the axis of the church, there are side arches separating the central nave from the chapels. These arches support a series of 0.09-m-thick domes, with a notably ellipsoidal shape, made of three solid brick layers. The church is enclosed to the north by a narthex wall, which has three openings at different levels, on the ground floor, the linteled entrance to the church; on the first floor, an arched entrance to the chancel; and, finally a rose window in the upper zone. Outside the church, the narthex has three levels (the last is the roof), all of them newly built to replace the previous ones.

Finally, the roof is completed by wooden trusses with iron ties and wooden studs. There are two types of truss. Type 1 corresponds to those supported by the main arches. In this case, brickwork pillar rests on the main arch and supports

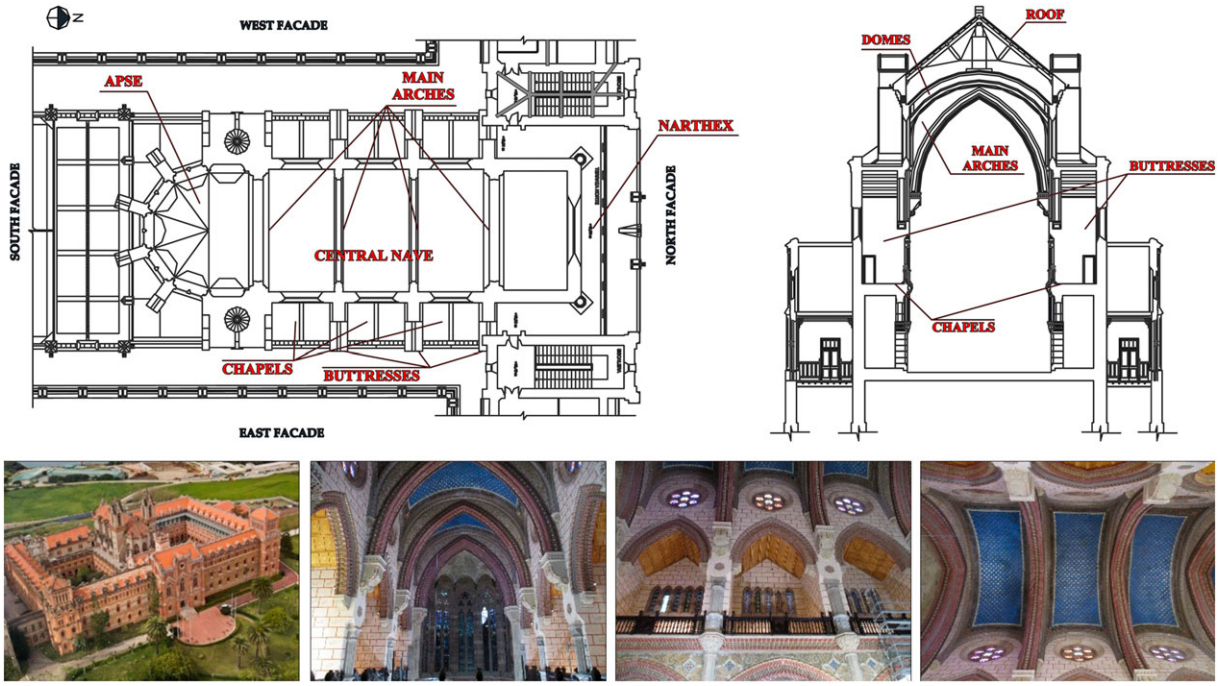


FIGURE 2 Building studied: Church of the Major Seminary of Comillas, Spain

the ridge beam. Thus, wooden props on the pillar support the rafters. Type 2 are trusses that, located between main arches, absorb the horizontal forces through iron ties. The line of the ties is polygonal with the aim of adapting their shape to the domes.

The building, of more than 100 years old, is built with mixed masonry of low-strength solid ceramic bricks and limestone pieces, with poor bonding. Throughout the rehabilitation work on the structure of the church, the presence of several rows of bricks with low-strength lime mortar in the lower part of the buttresses and walls of the building was detected.

3.1.2 | Structural problems

The building, which was in a relatively good state until the 1980s, underwent a progressive process of deterioration from when it fell into disuse. Consequently, as time passed, the church further deteriorated, which brought about the appearance of serious structural pathological processes affecting the building's stability[28] (deterioration of the wooden roof: damp, rotting, and loss of efficacy of the iron ties; excessive deformation of the wooden structure floors; significant cracking in the buttresses, facade walls, and domes; etc.).

As a safety measure, and owing to the significant size of the cracks observed in the domes of the building, in 2009, a propping system was positioned (Figure 3) that prevented their possible collapse, as well as the placement of tie-rods in the main arches to equilibrate the existing horizontal forces.

Additionally, in 2012, an exhaustive graphic and photographic record was made of the pathological processes occurring in the church,[29,30] with a total of 41 plans. The information was grouped according to the floorplan of the church, corresponding to the zones of arches and buttresses, domes, interior walls, and load-bearing exterior walls. The principal pathological processes detected during the inspection of the building are shown in Figure 4: (a) cracks, erosion, and detachment of the pieces forming the domes on the upper and lower faces; (b) damp, rot, and lack of cross section in wooden elements of the roof; (c) damp and efflorescence in walls and buttresses; (d) cracks and disaggregation of material in arches, walls, and buttresses; (e) large plants growing on the roof and walls of the building; and (f) damage due to biotic agents inside the church.

In the same way, a methodology for following up the damage detected was designed depending on its severity (Subsystem 1) and which would later enable the monitoring of possible movement of the structure during the future interventions.

3.1.3 | Principal interventions made

From 2014 to 2017, actuations were carried out to structurally consolidate the building along with those aimed at its integral artistic recuperation. Among the main structural actuations, the following can be highlighted: the injection



FIGURE 3 Photographs of the interior of the church after the installation of the propping system

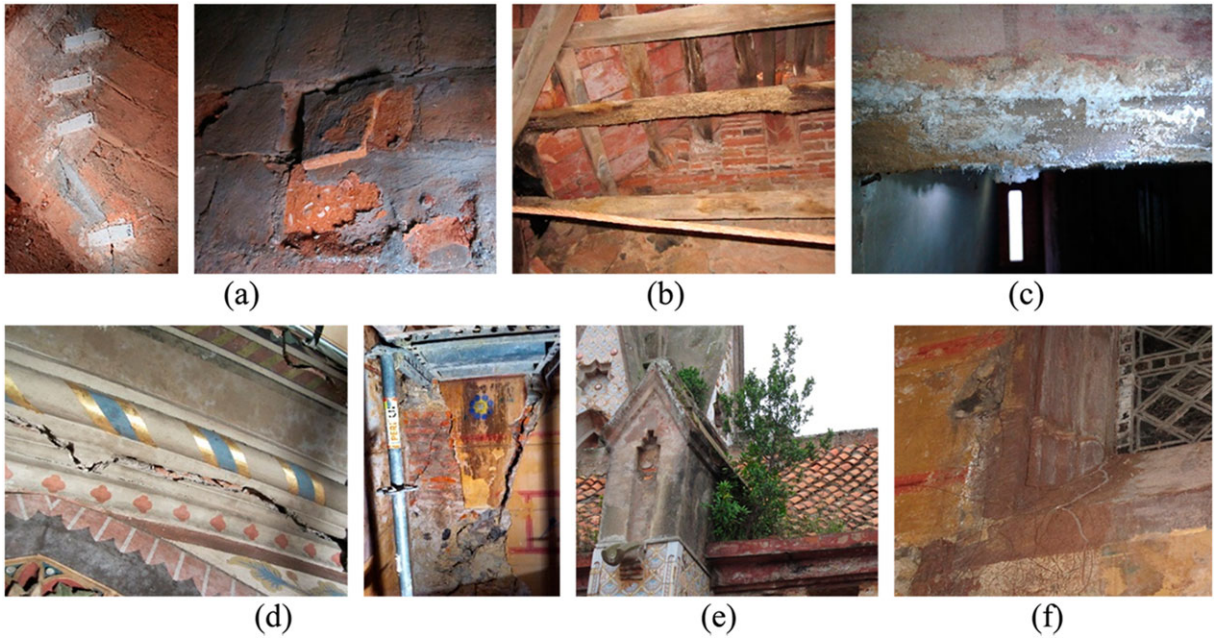


FIGURE 4 Principal pathological processes detected in the building

of arches and buttresses, the reinforcement of principal arches and domes, the interventions on the load-bearing walls adjacent to the two cloisters, and the replacement of the roof that led to an increase in the load on the structure.

Taking into account the component materials of the buttresses and the actuations that would be carried out on the building, it was necessary to verify the structural safety of the load-bearing elements with the aim of avoiding an increase in the stress levels in the existing masonry. In this sense, before the building rehabilitation process, a finite element analysis was done,[26] whose calculations were carried out in SAP2000.[31] Likewise, the analysis took into account the asymmetry of the building, given the difference in stiffness between the load-bearing wall-buttresses on the east side and those on the west side as a consequence of previous interventions.

Throughout the intervention process on the building, a detailed monitoring of the most relevant elements (arches, buttresses, domes, and walls) was done through the sensors implemented in the Subsystems 2–4. However, considering the mechanical characteristics of the materials and the magnitude of the intervention processes in the building, the article highlights the monitoring of the three most significant actuations[32,33]:

1. Execution of the passive reinforcement in the load-bearing east wall consisting of a reinforced concrete frame (Figure 5a), partially closing some gaps opened in an intervention previous to the monitoring period, so enabling the communication between the east cloister to the church. The equipment utilized for the monitoring of this actuation was included in Subsystem 2.
2. Removal of the propping system from the domes after reinforcing them (Figure 5b). Subsystem 3 includes the devices used for this purpose.
3. Intervention envisaged on the load-bearing west wall consisting of the aperture of gaps (approximately 4.00 × 4.60 m, including lintel and continuous footing) on the ground floor (Figure 5c) to facilitate communication between the west cloister and the church, and later reinforcement with concrete to guarantee the transmission of loads. The proposed monitoring elements are included in Subsystem 4.

3.2 | Instrumentation and data acquisition systems

The system configuration, with the possible implementation of new sensors at each stage (subsystems), enabled remote follow-up of any anomalous behaviour of the structure before, during, and after the interventions. Figure 6 shows a summary of the electronic sensors installed during the building monitoring process, which will be detailed in the following sections. In this respect, to aid understanding of the zones monitored by the component devices of each subsystem, the placement of these in the areas monitored will be detailed.



FIGURE 5 Sequence of actuation on the load-bearing walls and domes of the building: (a) east wall, (b) domes, (c) west wall, and (d) localization on floorplan

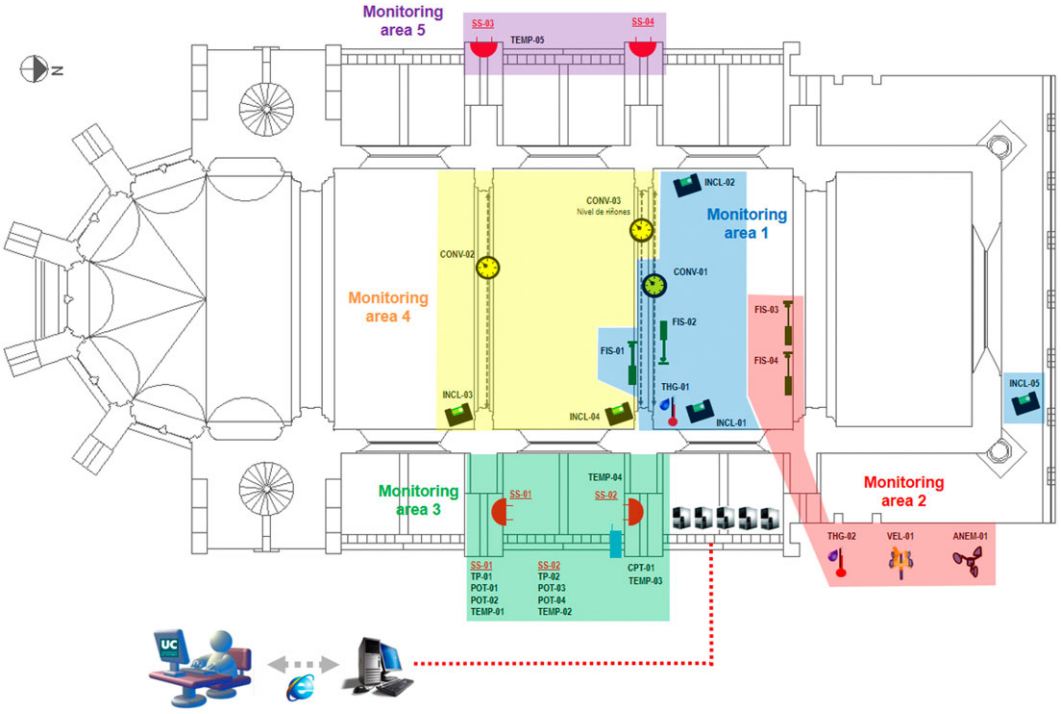


FIGURE 6 Scheme of distribution on the floorplan of the sensors installed in the monitoring areas

3.2.1 | Monitoring of damage and the first phase of actuations (Subsystem 1)

The structural responses, such as crack opening, convergence of load-bearing walls, inclination, and environmental conditions, were monitored according to the principal structural pathological processes detected in the building. For the follow-up, a total of 13 continuous electronic sensors were installed, distributed in two monitoring areas (Monitoring Areas 1 and 2, see Figure 6). The sensors of each zone were connected through to different remote data acquisition stations (data acquisition—Remote Terminal Units), and in turn, these were connected to an industrial PC, where the measurements were stored (data management and storage—Central Terminal Unit), and later managed via web (transmission and processing data).

Crackmeters used on arches and domes were installed at five locations of major damage. Servo-inclinometers were installed to perform monitoring at two locations in load-bearing exterior walls and on a wall of the narthex, with the aim of analysing their possible loss of verticality. Likewise, the convergence of load-bearing walls was measured with one cable extension sensor. Finally, two thermo-hygrometers, one anemometer, and one wind vane enabled the monitoring of environmental conditions inside and outside the church. Figure 7 shows some of these aforementioned sensors, and Table 1 provides their specifications.

Complementarily, for verification and additional measurement, 22 manual control elements were installed, of which 16 enabled the monitoring of aperture/closure of cracks (in arches and domes, in the latter on the upper and lower faces), and 6 monitored possible loss of verticality of walls. The configuration designed for evaluation of aperture/closure of cracks consisted of the placement of three monitoring points forming an equilateral triangle, so by measuring the distances among its vertices, the variation of the opening of the crack and the direction of displacement could be evaluated (Figure 8a). Reference discs were used as control points, and a DEMEC deformation meter (Mayes Instruments Limited) permitted the reading among them. As for the inclinations, these were measured on stainless steel inclinometer plates installed at strategically chosen localizations. The measurement of the inclinations was carried out using a portable ALPHA 4000 SX42891 digital inclinometer (Meggitt Sensorex) with a precision of two hundredths of a degree (Figure 8b).

3.2.2 | Equipment used to monitor the execution of the passive reinforcement on the east wall (Subsystem 2)

In the case of the passive reinforcement to be executed on the east load-bearing wall of the church, the objective was to analyse how the redistribution of loads between the buttresses and this reinforcement would be done. To carry out this analysis, it was not sufficient to monitor the conventional parameters already being monitored by Subsystem 1 (aperture/closure of cracks, inclinations, convergence between walls, and environmental conditions), which necessitated designing a subsystem of instrumentation that enabled continuous, real-time monitoring of the stress variations in Central Buttresses 2 and 3, which corresponded to the most loaded elements (Figure 5d), before, during, and after



FIGURE 7 Some of the sensors installed during damage monitoring and the first actuations

TABLE 1 Sensors installed in Monitoring Areas 1 and 2 integrated in Subsystem 1

Location	Monitoring element	Quantity	Model	Manufacturer
Monitoring Area 1	Crackmeter	2	LVDT V02505SAN3	Solartrom Metrology
	Servo-inclinometer	3	SX 41100	Sensorex
	Cable extension sensor	1	WS10-50-R1K-L10-SB0-D8-HG	ASM GmbH
	Thermo-hygrometer	1	PCE-PT18	PCE-Ibérica
Monitoring Area 2	Crackmeter	3	LVDT V02505SAN3	Solartrom Metrology
	Thermo-hygrometer	1	PCE-PT18	PCE-Ibérica
	Anemometer	1	3R KYWS	Darrera
	Wind vane	1	3R KYWD	Darrera



FIGURE 8 Follow-up of the manual control points: (a) aperture/closure of cracks and (b) variation of inclinations

the execution of the reinforcement. As a consequence, and given the versatility of the system designed, three devices were integrated into the system to monitor the stresses, two in Buttresses 2 and 3 and the third between the lintel of the reinforcement frame and the already existing lintel in the load-bearing wall installed in a previous intervention. This monitoring zone was denominated Monitoring Area 3 (Figure 6).

For the continuous follow-up of the stress variations in the buttresses, a device designed by our research group was used[32] that enabled the determination of initial service vertical stress of the structural masonry elements, and its later continuous monitoring over time. These were installed, as far as was practicable, in the zones with the greatest stresses (i.e., as close as possible to the foundation of the buttresses and toward their extremities). Meanwhile, a hydraulic pressure cell (200 × 400 mm) was installed on the reinforced concrete strengthening element, a sensor that is more widely used in civil infrastructures.[11,34,35] Thus, combining these devices, it was possible to monitor the execution and efficacy of the intervention.

Complementarily to these sensors, another 10 monitoring elements were installed. Six of them destined to the follow-up of displacements, with the aim of evaluating the relation existing between the stress variations and the displacements. Of these, four were displacement transducers (two on each buttress) for the remote, continuous monitoring and two manual control areas composed of three pairs of reference points. Of the four remaining sensors, three were destined to monitoring the thermal variations around each stress sensors and one to monitoring the environmental temperature in the test areas. These enabled the monitoring of the temperature around the fluid contained in the interior of the devices (stress sensors and hydraulic pressure cell), to analyse the possible influence of environmental temperature on the data recorded.

In summary, 11 continuous electronic sensors were implemented (Table 2) that connected to a new remote data acquisition station (Subsystem 2), and along with the two manual control areas, enabling the local monitoring of the cited structural elements. Figure 9 shows some of them placed on-site.

3.2.3 | Sensors used for monitoring the propping system removal of the domes (Subsystem 3)

After completing the actuations of the structural refurbishment of the brick domes and main arches,[33] the removal of the propping structure, supporting the elements under vertical actions, in the body of the church and the tie-rods in the

TABLE 2 Sensors installed in Monitoring Area 3

Location	Monitoring element	Quantity	Model	Manufacturer
Monitoring Area 3	Stress sensor	2	Designed by our research group	—
	Hydraulic pressure cell	1	0L111204000	Sisgeo
	Potentiometric displacement transducers	4	LM10/3M295K0M TO ES AW	RS Pro
	Surface temperature sensor	3	Thermocouple type T	TC Direct
	Thermo-hygrometer	1	PCE-P18	PCE-Ibérica

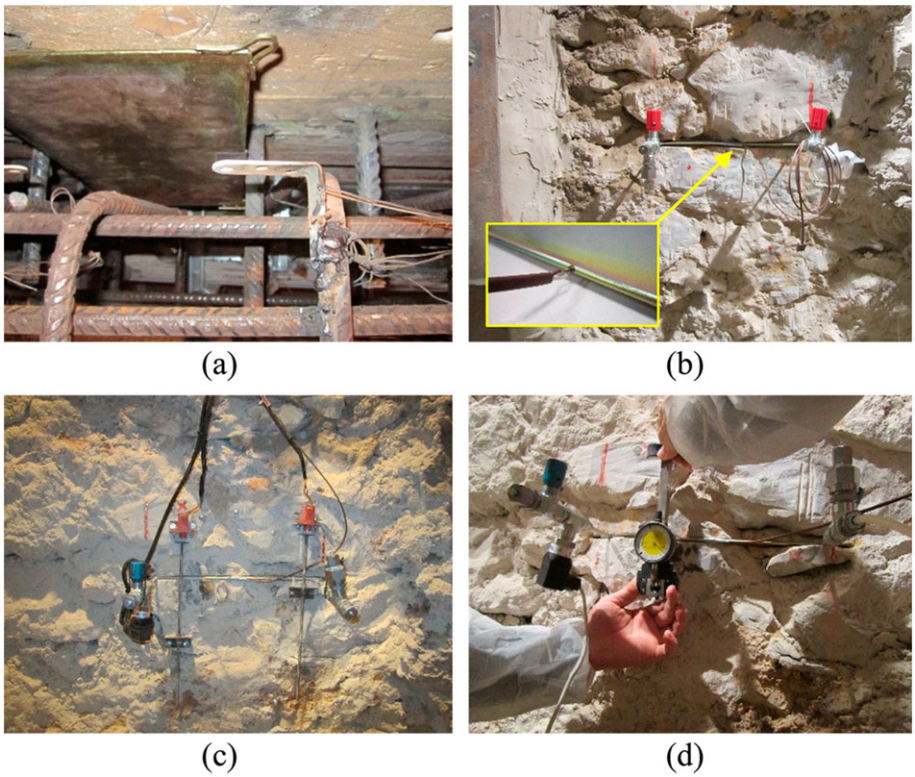


FIGURE 9 Examples of sensors deployed for the follow-up of the actuations: (a) hydraulic pressure cell in the upper part of the reinforcement lintel, (b) thermocouple installed in the stress sensor, (c) potentiometric displacement transducers, and (d) manual displacement record in one of the monitoring areas

main arches, to equilibrate the existing horizontal forces, was undertaken. As a safety measure, with the aim of evaluating the efficacy of the reinforcement installed, it was necessary to exhaustively monitor the displacements of the arches and the domes during this procedure. Therefore, a methodology was developed to enable the verification of the correct loading of the structural elements (gradually and not brusquely), as well as the detection, if necessary, of any anomaly occurring.

To do so, two monitoring systems were used. The first, a temporary system, was destined to monitoring the vertical displacements of the domes and horizontal aperture at the keystone of the arches. This follow-up, at a local level, was carried out using displacement transducers installed specifically for carrying out this actuation, which were connected to a temporary monitoring system. In summary, three displacement sensors were installed on each of the Domes 2, 3, and 4 of the church in the positions: keystone, right (west), and left (east); and one sensor on the keystone of each of the Arches 2 and 3. A list of the devices used is shown in Table 3.

The second monitoring system corresponded to the one installed permanently in the building in 2012. Given the versatility of the system, and taking into account the actuation to be monitored, four new sensors were added (Table 4) to the already mentioned ones. These, destined to Monitoring Area 4 (Figure 6) and implemented as Subsystem 3, were installed around the spring points of Arches 2 and 3, zones where the greatest horizontal stresses are produced. Additionally, before the prop removal process, the cable extension sensor located between the exterior walls in Monitoring

TABLE 3 Displacement transducers used in the prop removal of the domes, connected to a temporary monitoring system

Nomenclature	Displacement sensor	Location	Model	Manufacturer
Type 1	LVDT—50 mm	Keystone of dome	LVDT V02505SAN3	Solartrom Metrology
Type 2	Displacement transducer—25 mm	Keystone of arches	HLP190/SA1/25/1K	Penny & Giles
Type 3	Displacement transducer—100 mm	Toward the side of the domes	PM/100/5K	ELAP

TABLE 4 Sensors installed in Monitoring Area 4 and integrated into Subsystem 3

Location	Monitoring element	Quantity	Model	Manufacturer
Monitoring Area 4	Servo-inclinometer	2	SXI41900	Sensorex
	Cable extension sensor	2	WS10-50-R1K-L10-SB0-D8-HG	ASM GmbH

Area 1 was relocated at the approximate height of the reinforcing tie-rods between each opposing buttress of the church. Figure 10 illustrates the location of the new devices included in the permanent monitoring system of the building.

In addition to the mentioned devices, 10 new manual control points were added, four of them destined to the follow-up of possible loss of verticality (three inclinometer plates installed on buttresses and one on the narthex wall) and six to the monitoring of aperture/closure of cracks (elements located on the keystone and toward the side of Arches 2 and 3). These elements provided a greater number of monitoring points in the building.

3.2.4 | Description of the proposed sensors for monitoring the intervention on the west wall (Subsystem 4)

The intervention envisaged on the west load-bearing exterior wall of the church consisted of the opening of gaps on the ground floor to facilitate communication between the cloisters and the church. To do so, the transmission of loads was guaranteed through the execution of a reinforced concrete frame with interior pillars. As a consequence, and given the magnitude of the intervention to be carried out, it was necessary to monitor the evolution of the state of stresses in the buttresses on the west side during the wall cutting work and the later execution of the concrete reinforcement.

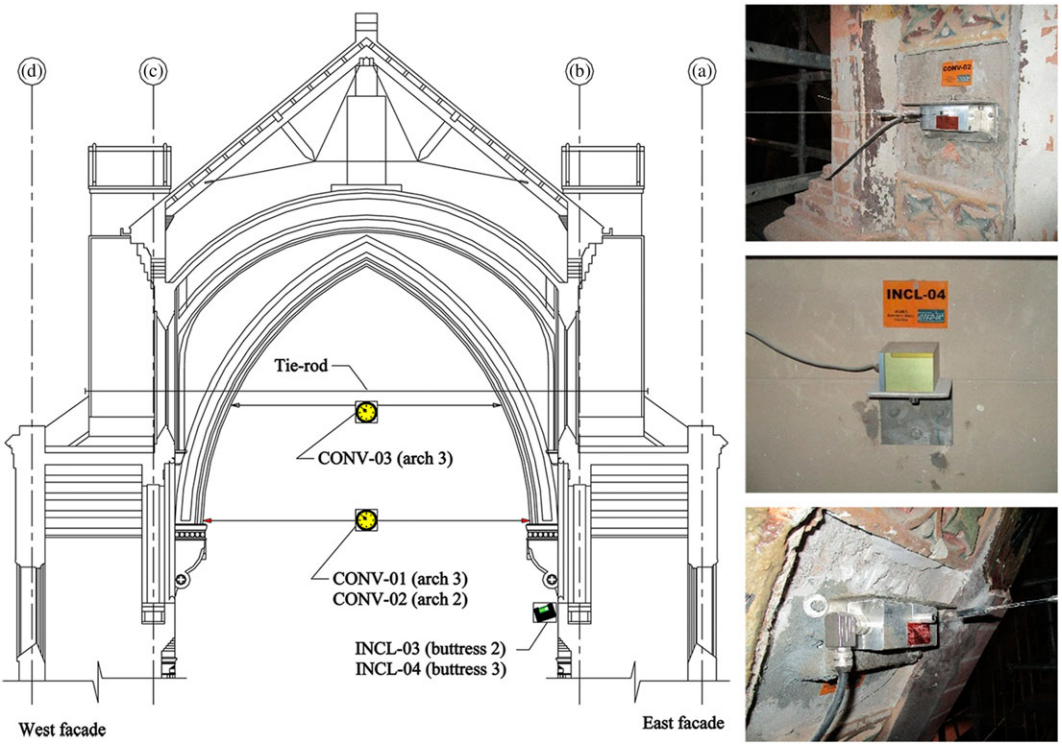


FIGURE 10 Location on the elevation of the four new sensors installed in Monitoring Area 4

For this purpose, two stress sensors were added with similar characteristics to those used on the east buttresses. These were installed on Central Buttresses 2 and 3 (Monitoring Area 5, Figure 6), with the aim of determining the initial service stress existing in the buttresses (Figure 11a) and monitoring the variations registered during the cutting process and the execution of the reinforcement (Figure 11b).

Additionally, and in a similar way to the opposing buttresses, an environmental temperature sensor was added to analyse the effect of environmental factors on the measurements, and manual control elements, composed of four pairs of reference points, were introduced to follow-up the displacements, providing elements to contrast with the stress variations (Table 5).

3.2.5 | Data acquisition and software development

The integration of the system was based on the implementation of Remote Terminal Unit (RTU) architecture in an industrial PC acting as a master (Master Terminal Unit, MTU).[36,37] This, along with suitable data acquisition cards for each sensor type, carried out uninterrupted data collection. Additionally, alongside the installation of an application server, which periodically communicated with the system, there was remote, real-time access to the data stored in a database. The system designed also incorporated actuation protocols that systematically evaluated the sensor records in the computer, automatically defining alerts if the magnitudes being monitored exceeded specific prefixed limits. Thus, when receiving an excessive value, an instantaneous alert would be sent via email to the assigned addresses. Finally, a web server provided remote access to both the data themselves and the system configuration, using a novel application developed in JavaFX, a platform for developing rich Internet applications.

The system developed ad hoc in this research utilizes user interface software that can efficiently manage the received data and enable various administrators to check the data easily. Software access levels can be set to limit the range of activity for each administrator. In this respect, Figure 12 shows a visualization of the JavaFX application developed.

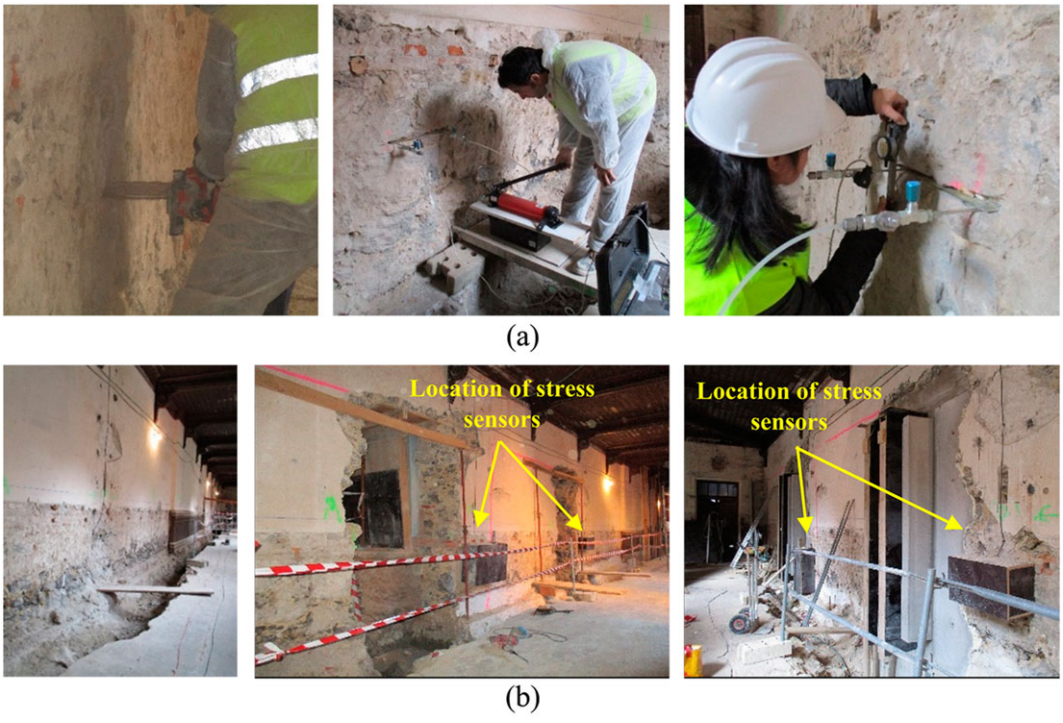


FIGURE 11 (a) Determination of initial service stress in the masonry and (b) continuous monitoring of the stress variations before, during, and after the intervention process

TABLE 5 Sensors installed in Monitoring Area 5

Location	Monitoring element	Quantity	Model	Manufacturer
Monitoring Area 5	Stress sensor	2	Designed by our research group	—
	Thermo-hygrometer	1	PCE-P18	PCE-Ibérica

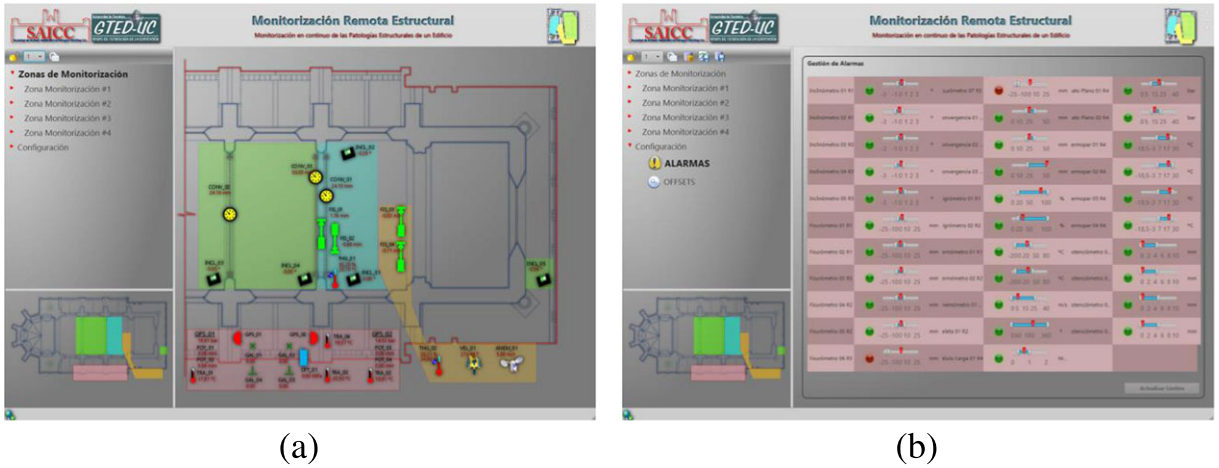


FIGURE 12 User interface software program: (a) real-time data monitoring and (b) defined automatic alerts

4 | ANALYSIS OF THE DATA FROM THE INTEGRATED SYSTEM

The analysis of the monitoring data from the integrated system was carried out over a period of 5 years, from October 2012 to September 2017. The main goals were to identify the occurrence of possible unstable phenomena or damage mechanisms during the monitoring of the pathological processes and to provide follow-up of the behaviour of the structural elements during the different interventions carried out on the building. The most relevant results are shown next.

In relation to the crack aperture/closure, it was found that the seasonal temperature variations imply opening or closing cycles of the cracks in the church. In general, the cracks exhibited a maximum opening in winter–autumn and a closing process during the summer. As an example, the data acquired by crackmeter FIS-01 from October 2012 to September 2017 are shown in Figure 13. During the period from October 2012 to January 2015, corresponding to the follow-up of the pathological processes and the first working actuations, the crack openings in Arch 3 were checked, and the measurements recorded were of the order of hundredths of millimetres. Similar behaviour was observed in the homologous sensors installed in the arches under the domes.

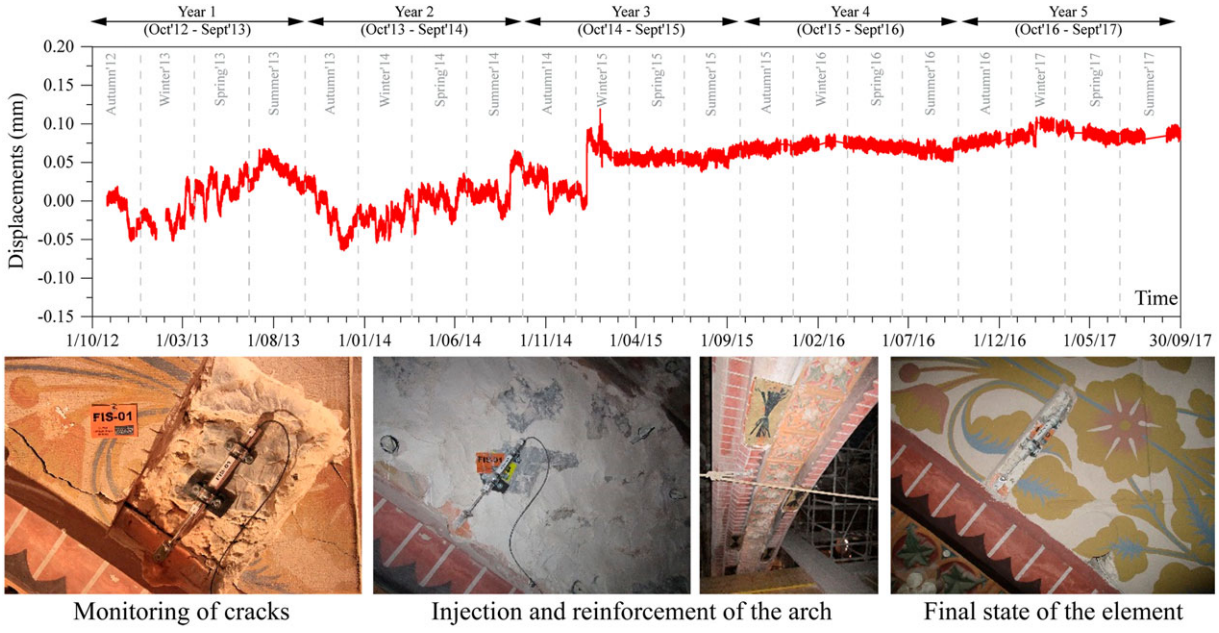


FIGURE 13 Evolution registered of crack width recorded by crackmeter FIS-01 in Arch 3 during 5 years of monitoring ("-" indicates crack opening)

Meanwhile, crackmeters FIS-04 and FIS-05 located on the domes (below and above, respectively) registered higher amplitudes than the cited ones, maximum values of 0.71 and 0.58 mm, respectively, in both cases, as a consequence of being installed on a less stiff element, whereas the records of FIS-05 were additionally affected by the direct effect of exterior atmospheric changes.[29] In February 2015, the consolidation work on the arches and domes was started (injection of cracks and reinforcement), which on finalizing, were reflected in lesser variability of the records of the sensors. After these interventions, the amplitudes presented by the crackmeters showed a stable tendency as can be seen in Figure 13, corresponding to FIS-01.

In relation to the on-site monitoring of inclinations, the data recorded by the servo-inclinometers between 2012 and 2017 show very small angular variations. For instance, Figure 14 depicts the time history of sensor INCL-01. In a similar way to the case of crack widening, the evolution of inclination recorded appears to be related to the temperature fluctuations registered. Additionally, it should be remarked that the effects of wind actions on the behaviour recorded by the servo-inclinometers could be observed. Thus, greater values of inclination were measured when significant wind speeds were recorded. Figure 14 shows the wind magnitudes recorded from October 2013 to September 2014, which had repercussions for the sensor records.

According to the methodology envisaged and the results obtained, the manual control elements provided contrast measurements to those recorded by the continuous sensors. A close-up of the period between February 2013 and April 2014 of crackmeter FIS-05 and manual control point G-03 is displayed in Figure 15a. Good correlation can be observed between the aperture/closure of cracks determined through the two methodologies. Similarly, the results measured by the inclinometer plates provided equivalent results to those of the servo-inclinometers installed adjacent to them. As an

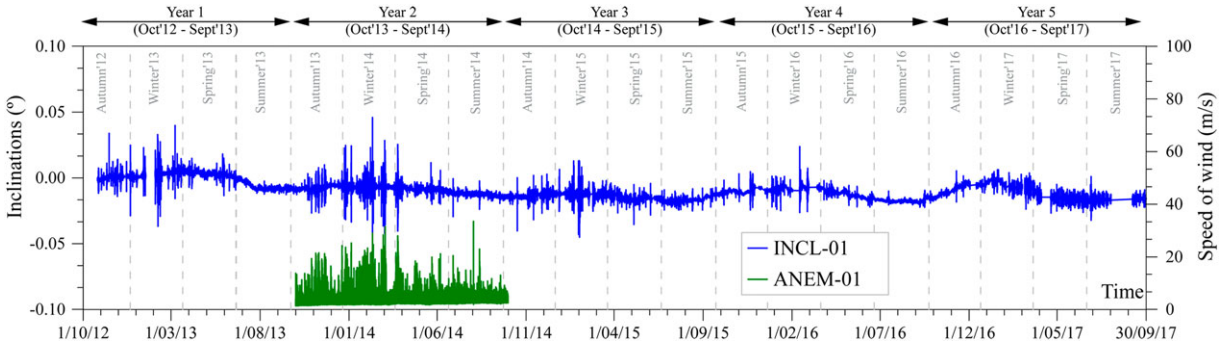


FIGURE 14 Evolution registered of the inclinations recorded by servo-inclinometer INCL-01 located on the east load-bearing wall during 5 years of monitoring ("-" indicates inclination toward the outside of the wall)

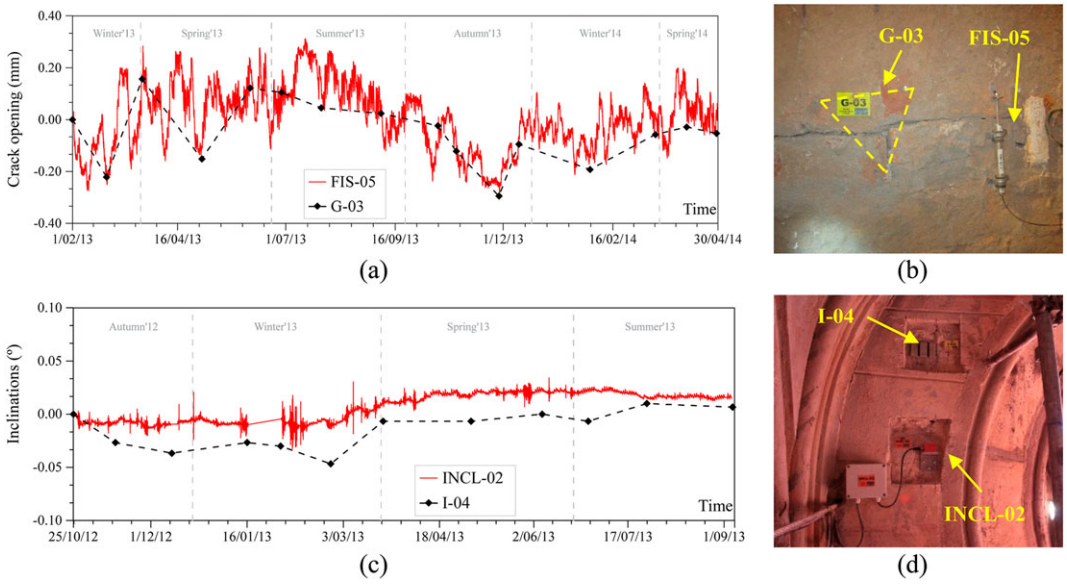


FIGURE 15 (a and b) Opening of cracks registered by FIS-05 emplaced in the extrados of a dome and comparison with the measurement obtained manually by G-03 installed adjacent to it ("-" indicates opening of cracks) and (c and d) inclinations registered by INCL-02 located on the east load-bearing wall and comparison with the measurement obtained manually with I-04 ("-" indicates loss of verticality toward the outside of the wall)

example, Figure 15c shows the records of inclinometer plate I-04 and by servo-inclinometer INCL-02. The small variations observed, fundamentally in the inclination measurements, could be associated with the difference in precision of the two monitoring systems. Thus, having validated these results, the manual control points enabled additional monitoring and follow-up of a larger number of structural elements of the building.

In order to illustrate the potential of the system designed, next the evolution of the records is shown relative to the stresses and displacements (obtained through discrete in situ monitoring and by continuous remote monitoring), before, during, and after the execution of the reinforcement of the east load-bearing wall. Additionally, analysis was done of the correlation existing between the two parameters (stress and displacement), as well as the influence of the thermal gradient measurements on the stresses. Detailed analysis of the devices mentioned and the sensors installed in Subsystem 1 provided a local and global response of the structure during the intervention process.

Figure 16 illustrates the evolution of the stresses in Buttresses 2 and 3 located in the east of the building (SS-01 and SS-02) during 23 months of the follow-up. The records started in December 2015 after determining the initial stress state of the buttresses, discontinuous reference also included in the figure (SS-01-I and SS-02-I). In relation to the evolution of stresses over the period monitored, it should be said that the variations recorded were not significant, and they were influenced to a great extent by the environmental thermal variations of the building. Figure 16 illustrates, as well as the stress evolution, the values recorded by the thermocouples installed in the stress sensors (TEMP-01 and TEMP-02). Behaviour can be observed in accordance with the seasonal variations, decreasing stress values being recorded with reducing temperatures, and vice versa. Thus, in the period from December 2015 to October 2017, the variations recorded ranged from 1.43 to 1.88 MPa in the sensor installed on Buttress 2 (SS-01) and 1.05 to 1.33 MPa in the other one on Buttress 3 (SS-02).

With respect to the initial behaviour of the sensors, it should be indicated that the difference between the initial stress taken as a reference during the continuous follow-up and the service stress value determined corresponded to the tasks of over-pressurization of the pressure pad.[32] After this actuation, the behaviour observed in the devices in the first stages showed a slight downward slope associated with the readjustment of the system "pressure pad—masonry structure" under maintained pressure.

The behaviour of the stresses and displacements in the monitored areas on the buttresses showed good correlation. As the pressure increased, opening was noted by the transducers. As an example, Figure 17a illustrates the behaviour shown by the devices installed on Buttress 2. As for the follow-up of the displacements measured through the two methodologies (manual and continuous), analogous behaviour was also observed between the two parameters. As an example, in Figure 17b, the evolution of the movements can be observed between manual control point D-01 and the displacement transducers POT-01 and POT-02. As can be seen, in the mentioned period (7 months), the fit of the measurements is good.

Consequently, the displacement records enabled the evolution of the stresses in the masonry to be contrasted, and the adequate fit between them to be verified, so making an extra monitoring element available with the aim of detecting possible measurement errors and/or problems of pressure loss.

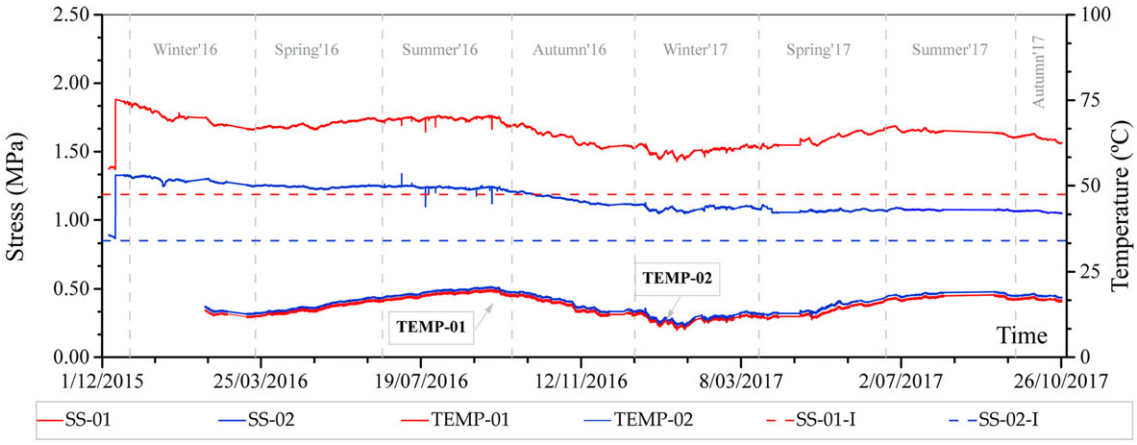


FIGURE 16 Graph of evolution of stresses SS-01 and SS-02 on East Buttresses 2 and 3 (Monitoring Area 3) and constant discontinuous reference of the initial stress value determined in the masonry (SS-01-I and SS-02-I). In addition, the temperature by the thermocouples (TEMP-01 and TEMP-02) installed in the sensors was recorded with the aim of assessing the influence of temperature variations on the measurements

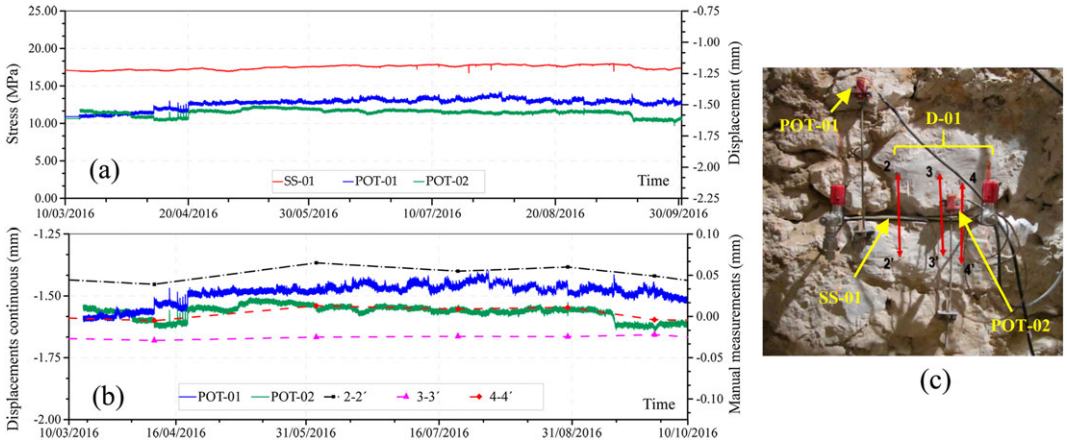


FIGURE 17 Examples of evolution graphs registered on Buttress 2 ("+" indicates aperture of the crack and increase in pressure): (a) evolution of the pressures (SS-01) and the displacements (POT-01 and POT-02); (b) records of the continuous sensors (POT-01 and POT-02) and comparison with the manual one (D-01); and (c) location on the projection of the control elements

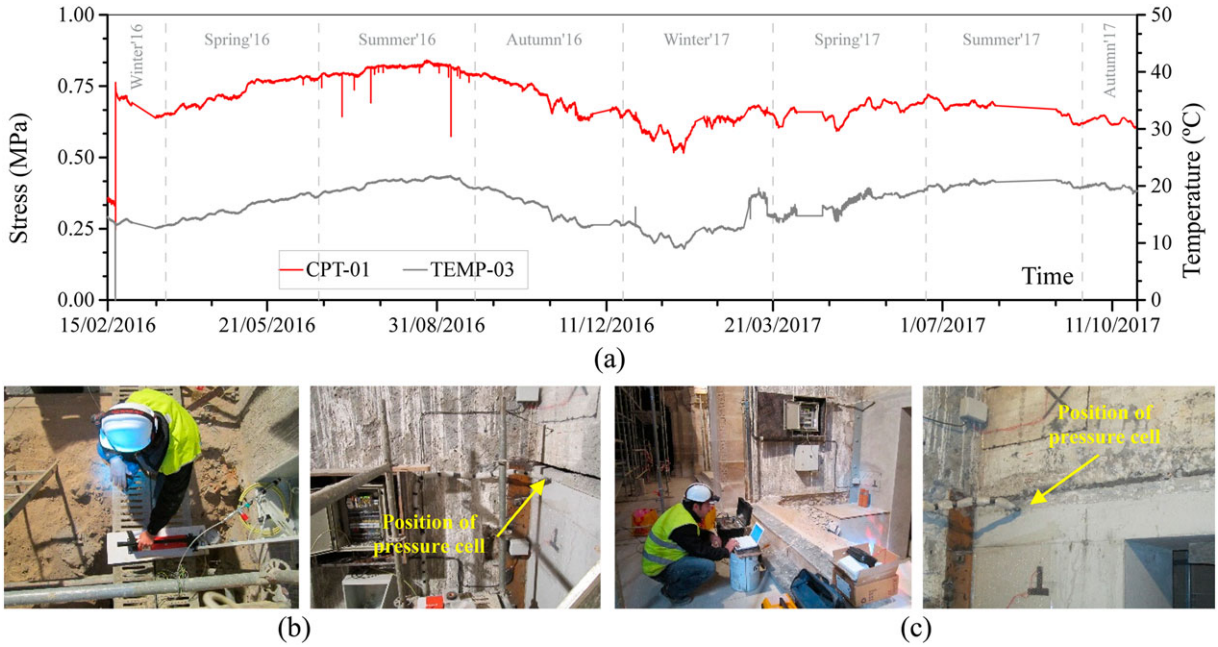


FIGURE 18 Evolution of the pressures (CPT-01) in the pressure cell installed on one of the reinforced concrete supports of the central span (east wall) and temperature recorded by the thermocouple installed in the sensor (TEMP-03)

The monitoring of the pressure cell (CPT-01) from February 2016 to October 2017 is shown in Figure 18a. As for the behaviour displayed since the execution of the concrete reinforcement, this sensor has recorded slight fluctuations with a similar tendency to the thermal variations registered by the thermocouple adjacent to it (TEMP-03). From June to October 2016, the value of 0.75 MPa was surpassed, which was not reached in the subsequent months, despite maintaining a similar tendency to the behaviour displayed in the same period of the previous year.

In relation to the values recorded during the monitored period, it should be said that initially, a stress of 0.375 MPa was introduced (after the concreting actuations, Figure 18b), later repressurizing the system up to 0.763 MPa once the last phase of the passive reinforcement had been done (a value that constituted the starting reference pressure level), Figure 18c. At the end of February 2016, the value of this parameter was 0.693 MPa (which can be explained by the process of deformation of the device under sustained pressure, which brought about a relaxation of the internal pressure).

During the execution of the reinforcement of the east wall (December 2015 to January 2016), the sensors installed in Subsystem 1 were also followed up (Monitoring Areas 1 and 2), which enabled the global monitoring of the building. It should be noted that these sensors did not display any anomalous behaviour during their follow-up. As an example,

Figures 13 and 14 show that the values recorded by FIS-01 and INCL-01 demonstrate that the intervention did not affect the rest of the elements of the structure.

According to the methodology adopted for the monitoring of the propping system removal and slackening of the tie-rods in the domes carried out at the end of February 2016, the vertical displacement in domes and horizontal aperture in keystones of arches were monitored using a series of sensors installed temporarily for this purpose. In this regard, it should be said that no anomalous behaviour was registered during the operation.[33] Moreover, most of the variations were stabilized during the days after the operation.

The evolution of the movements registered by the system of remote continuous monitoring sensors (Subsystem 1–3) showed little variation during the actuation. Before the intervention (between 2012 and 2016), there were no significant movements recorded, mostly corresponding to the thermal gradients present.

As an example, Figure 19 shows the evolution graphs of some of the devices from the day before the propping system removal until 7 months afterwards. The crackmeters maintained relatively stable behaviour during the monitored period. Figure 13 shows the behaviour of FIS-01. Only FIS04 registered a slight variation (Figure 19a), which was associated with the instant in which the props close to the sensor were removed (Figure 19b). However, it should be said that the variation was instantaneous and insignificant. It can be seen in the graph that during following 7 months, the recorded values remained relatively constant.

As for the relative displacements of the cable extension sensors, these did not undergo significant changes. Thus, Figure 19c shows, as an example, the monitoring of sensor CONV-03 installed at the height of the tie-rods (Figures 10 and 19d). During the monitored period, before, during, and after the propping system removal actuation, the values recorded remained around the origin. The slight variation registered during the months after the actuation was a consequence of the work close to the sensor.

The behaviour of the servo-inclinometers during the actuation monitored did not show abnormal angular variation. Figure 14 shows the behaviour maintained by sensor INCL-01. Finally, it should be indicated that the variations recorded by the stress devices on the east buttresses and by the pressure cell installed on the concrete reinforcement were not significant (Figures 16 and 18, February 2016).

The variation of the stress state in the west buttresses was monitored locally by stress sensors SS-03 and SS-04 during the opening of the gaps in the west load-bearing wall on the ground floor. Moreover, a temperature sensor (TEMP-05) installed in monitoring area enabled the evaluation of the environmental influence on them. These devices along with the rest of the previously installed sensors (Subsystems 1–4) enabled local and global follow-up of the structure.

As for the evolution of the stresses in Buttresses 2 and 3 on the west side, similar behaviour to the opposite counterparts in Monitoring Area 3 on the east side was observed. In a general sense, the sensors displayed behaviour in

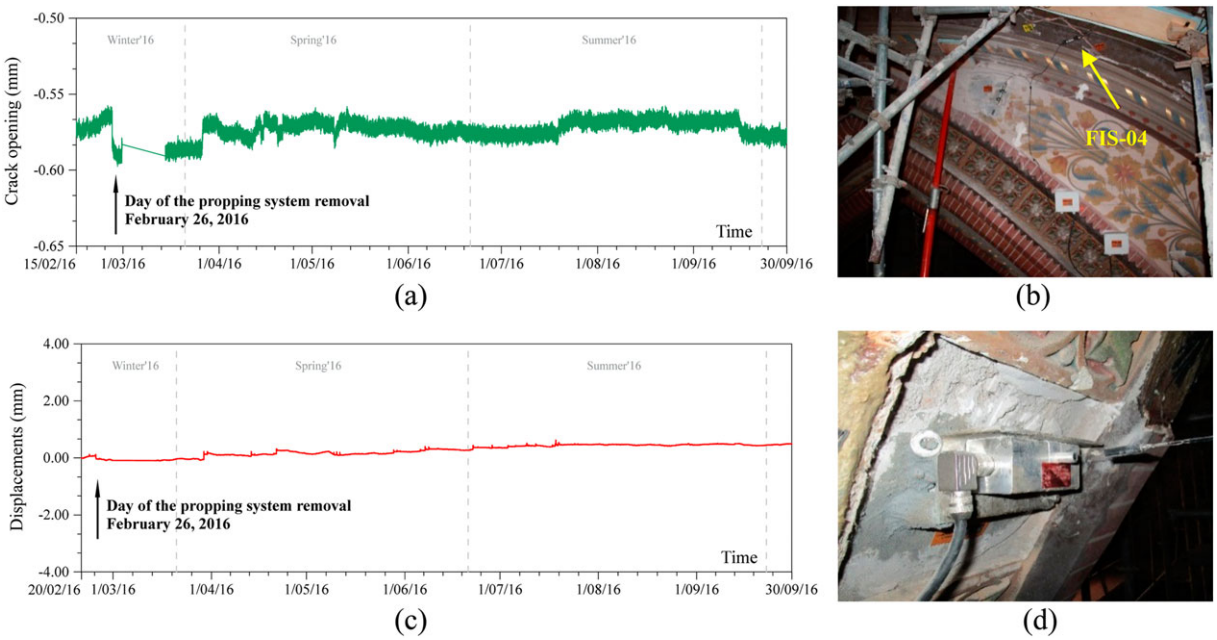


FIGURE 19 (a) Evolution and (b) location of crackmeter FIS-04 in Dome 4—lower face ("-" indicates crack opening) and (c) displacements recorded by cable extension sensor CONV-03 (d) located at the height of the tie-road ("-" indicates approximation of the walls)

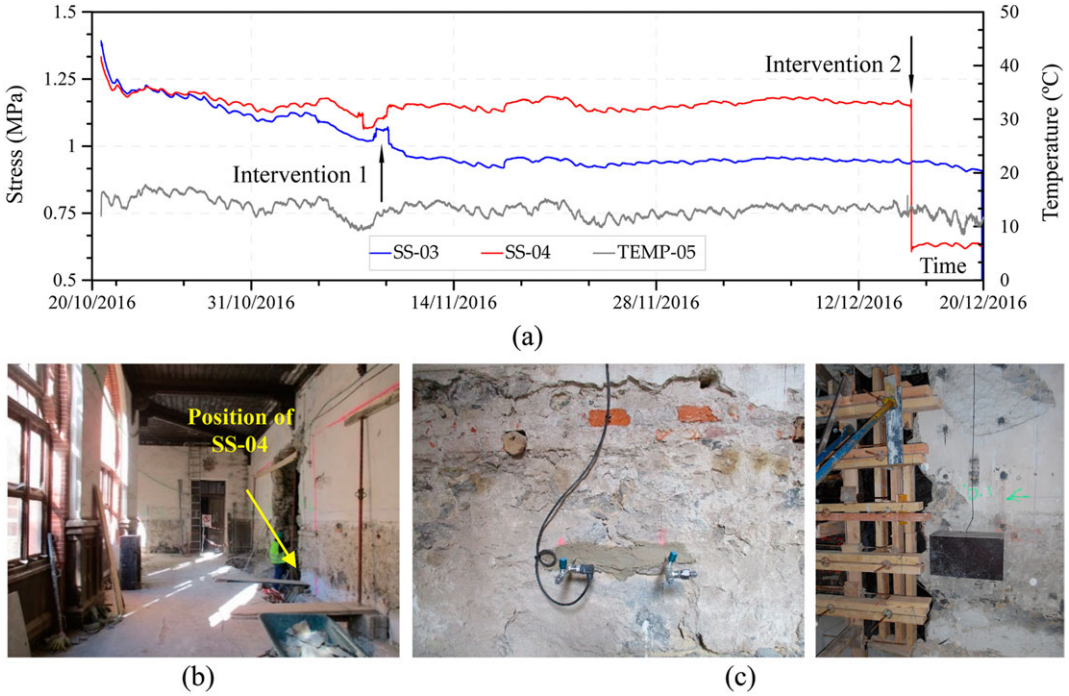


FIGURE 20 (a) Graph of evolution of stress variations in West Buttresses 2 and 3 (SS-03 and SS-04) and temperature register (TEMP-05); (b) incidence of the sun on the west load-bearing wall; and (c) substrate of device SS-04 before and after Intervention 2

consonance with the seasonal variations in temperature, confirming the existence of a high level of correlation between the two parameters. Figure 20a depicts the data recorded between October to December 2016 by sensors SS-03, SS-04, and TEMP-05. It should be remarked that these sensors were more influenced than the other sensors installed in the building by the diurnal and nocturnal thermal cycles (daily cycles) because these sensors were located in a cloister whose carpentry had not been restored, with part of the glasswork broken or inexistent, so there was a greater climatic incidence (Figure 20b).

As was mentioned before, the stress sensors enabled the monitoring of each of the interventions being carried out on the walls. As an example, two of the actuations were recorded in the graph in Figure 20a. The first (Intervention 1) reflected the start of the work related to the opening of gaps towards the west cloister on November 4th, 2016, when a variation was detected in the stresses in West Buttress 2 (SS-03). The second intervention recorded corresponds to the work opening the gap adjacent to device SS-04 located on West Buttress 3. In this case, an abrupt and significant decrease in the stress was detected (of around 0.6 MPa). This behaviour was a consequence of the process of perforation near the sensor installed on the buttress (Figure 20c) and placement of the structural steel of the adjacent column.

As well as the continuous stress monitoring points, a follow-up was done of the evolution of the displacements recorded manually in the monitoring areas (D-03 and D-04). As an example, Figure 21a illustrates the follow-up of West Buttress 3 of the building (D-04). It can be seen that the values mostly recorded small variations over time and around the origin. The specific significant variations displayed by the manual control points were associated with the

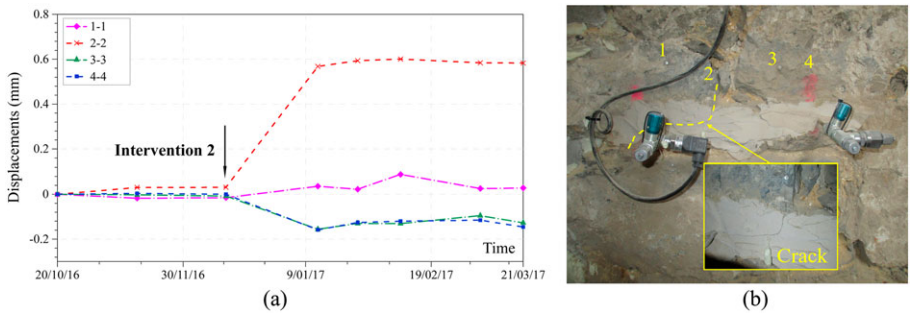


FIGURE 21 (a) Evolution of the displacements at manual control points D-03 located on West Buttress 3 of the building ("+" indicates increase in the control distance); (b) aperture of the crack near Monitoring Point 2 of the stress sensor

actuations carried out near the stress sensors, which consequently confirmed the relation between the evolution of displacements and stresses.

Of these, the correlation should be highlighted between the abrupt decrease in stress in sensor SS-04, as a consequence of Intervention 2, and the evolution of the displacements at some points in monitoring area D-4. It can be seen that as a result of this actuation, a crack was generated in the monitoring area that provoked a modification in the associated displacement records. Control point 2–2′, which registered a significant increment, was affected to a greater extent, as it was closer to the crack (Figure 21b).

5 | CONCLUSIONS

This research developed an integrated SHM system that was applied to a historic masonry building (Modernist Church in Comillas, Spain). This was a modular, easily adaptable system that enabled the modification, elimination, or addition of new devices and data acquisition systems according to the necessities of areas and/or elements to be monitored before, during, and after the interventions carried out. Thus, a system on the basis of the measured global and local responses of the structure during the monitored period was made available, ensuring the safety and stability of interventions on the church.

The long-term (5 years) monitoring results recorded from a large number of sensors (traditional and new technologies applied to the sector of rehabilitation of historic buildings) demonstrated the capacity and versatility of the system to monitor the different structural reinforcement interventions. These were carried out without the occurrence of any irregularity during the development of the work.

From the analyses carried out, it can be concluded that there is high correlation between the registers of the sensors and the temperature variation.

It was found that the registers (crack amplitudes, inclinations, and displacements) using manual follow-up and continuous electronic monitoring showed a good fit. The small differences in the graphs may be associated with the difference in precision of the two measurement systems. Therefore, it is recommended that the two monitoring methods are combined, not only to provide contrast measurements but also to increase the number of monitoring points at a reasonable cost.

The new methodology implemented on several buttresses of the church for continuous monitoring of the stress variation was extremely useful, given that it enabled the interventions carried out on the building to be monitored without any occurrence of significant anomalies. The service stresses were maintained below the masonry's strength limits.

Finally, it can be concluded that the SHM system designed played an important role in the structural analysis. Its use before, during, and after the intervention contributes to an increase in the safety of the building and of the personnel involved in the intervention, as it provides alerts about the possible danger caused by damage and/or degraded elements that are being monitored. This is a very interesting tool for application during interventions in historic buildings.

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